

(Assignment-3)

Q.1:- Explain the concept of situation calculus?

Ans:- Situation calculus is a formal logical framework used to represent and reason about actions, change, and knowledge in a dynamic world.

It allows for describing scenarios involving multiple agents, their actions and the effects of their actions.

Q.2:- what is knowledge representation, define its concept?

Ans:- Knowledge representation is the process of encoding and structuring information in a way that can be understood and processed by intelligent systems, such as computers or AI models. It involves capturing knowledge about the world, organizing it into a structured format, and providing mechanisms for reasoning and inference.

↳ The concept of knowledge representation revolves around finding suitable ways to represent and store different types of knowledge, including facts, concepts, rules and procedures.

Q.3:- what is Bayesian network? Explain its concept.

Ans:- A Bayesian network, also known as belief network or probabilistic graphical model, is a graphical representation of probabilistic

relationships among a set of variables. It is based on Bayesian probability relationships among a set of variables.

It is based on Bayesian probability theory and provides a compact and intuitive way to model and reason about uncertain or probabilistic knowledge.

↳ The concept of Bayesian network involves two main components:

- (i) nodes and
- (ii) edges.

Nodes represent variables, which can be discrete or continuous, and edges represent probabilistic dependencies or relationships between variables.

Q.4: → what is fuzzy logic? Explain with suitable example.

Ans: → fuzzy logic is a mathematical framework that deals with reasoning and decision making in situations where uncertainty, imprecision, or vagueness is present. It provides a way to handle and manipulate fuzzy or partially defined concepts, allowing for more flexible and reason compared to traditional binary logic.

An example, that illustrates fuzzy logic is the control of a room's air conditioning system. In a traditional binary logic, the temperature in the room would be classified as either "hot" or "cold" based on a fixed threshold value.

Q.5: → Differentiate between propositional logic and predicate logic.

propositional logic	predicate logic
1.) It is basic and most widely used logic. Also known as Boolean logic.	It is an extension of propositional logic covering predicates and quantification.
2.) A proposition has a specific truth value, either true or false.	A predicate's truth value depends on the variable value.
3.) It cannot deal with sets of entities	It can deal with set of entities with the help of quantifiers.
4.) It is more generalized representation.	It is a more specialized representation.